



Data Management, Privacy, and Ethics

CSIT205 Generative AI – 2025



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Wollongong
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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*

Dr Andrew Yoong, SIM



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, partially cut off. The 'U' is at the top, the 'O' is in the middle, and the 'W' is at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
community.

Find out more, seek support or join
the UOW Ally Network
uow.info/ally

Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

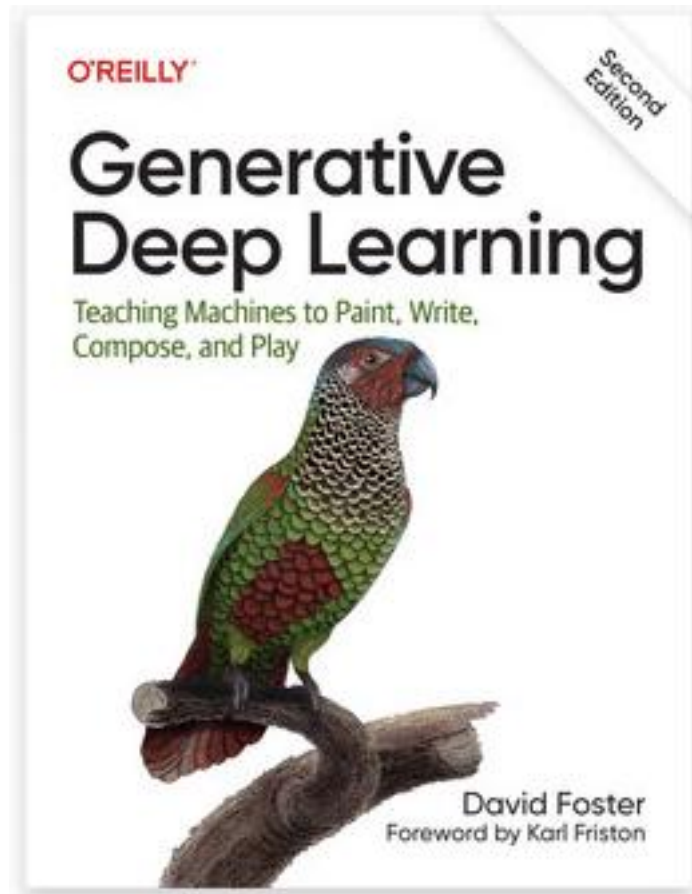
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

What did we discuss last week?



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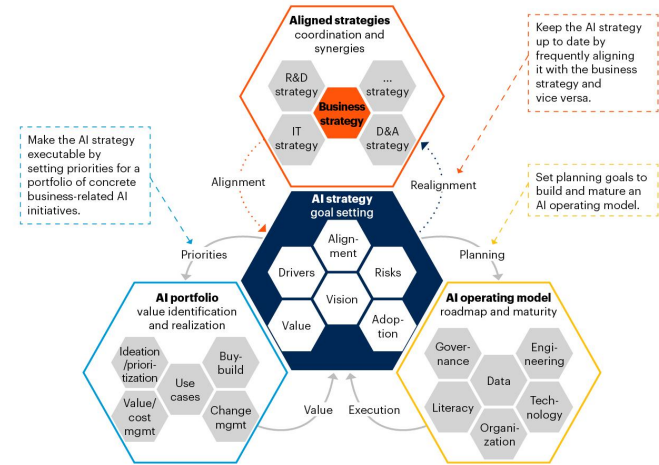
The AI Strategy

Owned by Head of AI / Data depending on company size and responsibilities

Interfaces with:

- *Aligned Strategies: continuous re-alignment*
- *AI Portfolio: generates value and receives priorities*
- *Operating Model: receives planning and executes AI Strategy*

The AI Strategy in Context



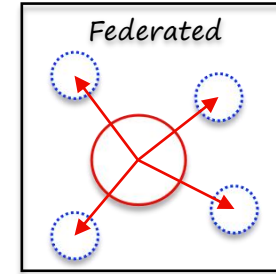
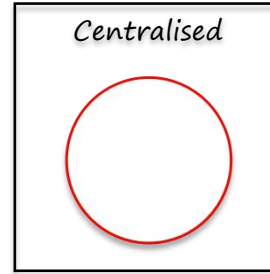
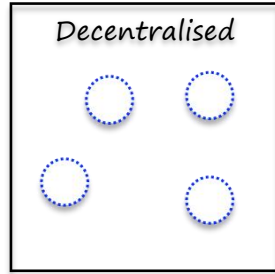
Source: Gartner
805859_C

Gartner

Den Hamer & Bhati, 2024

Centralised vs Decentralised

How are you going to organise for value delivery?



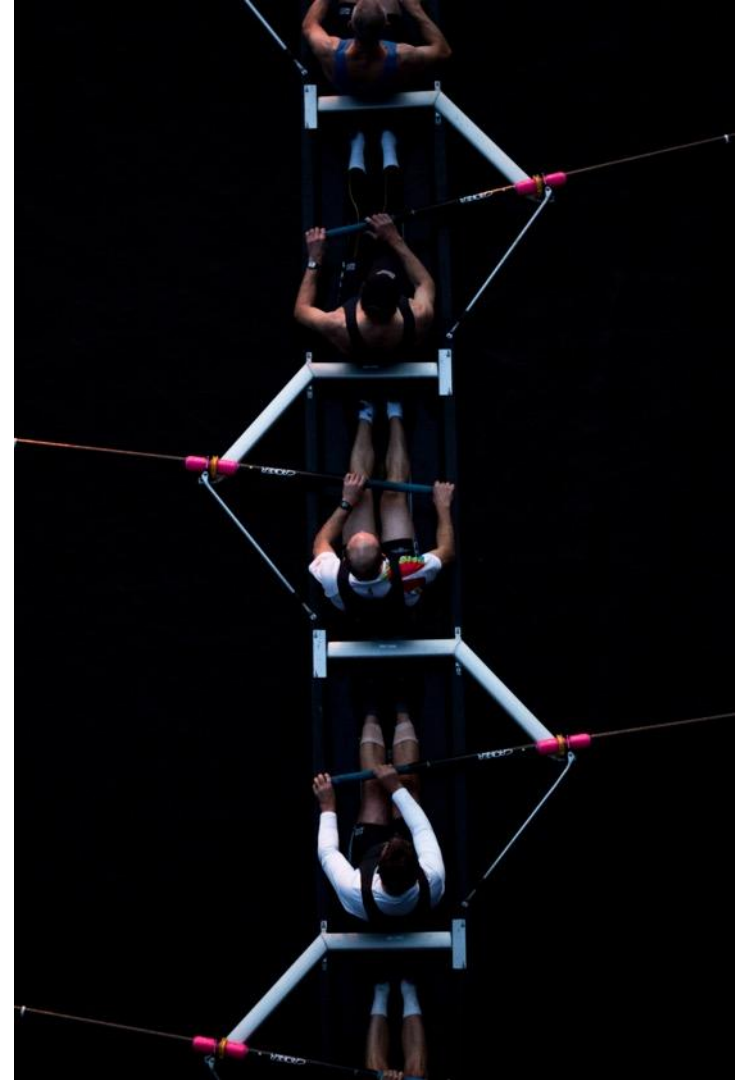
Time-to-market	Fast	Slow	Increasingly fast due to standardisation
Tools / tech	Risk of duplication	One set of tools / tech	Flexible' set of tools / tech
Governance	Challenging to control	Top-down control	Controlled through standards and guidance
Communication	Fast within teams	Slow comms with business	Closer ties to business
Small/large orgs	Can work for large organisations	Can work for small organisations	Can work for large organisations
Decision making	Fast, autonomous	Slower, top-down (can be positive!)	Top-down and autonomous
Development opportunities	Good	Fewer	Varying development opportunities: Centre-of-excellence can drive development quickly

Why are we here?

CSIT205 Generative AI - Module 3

Data Management, Privacy, and Ethics

- *AI Operating Model (Part 2)*
- *Data - Privacy*
- *Governance - Ethics and Responsibility*



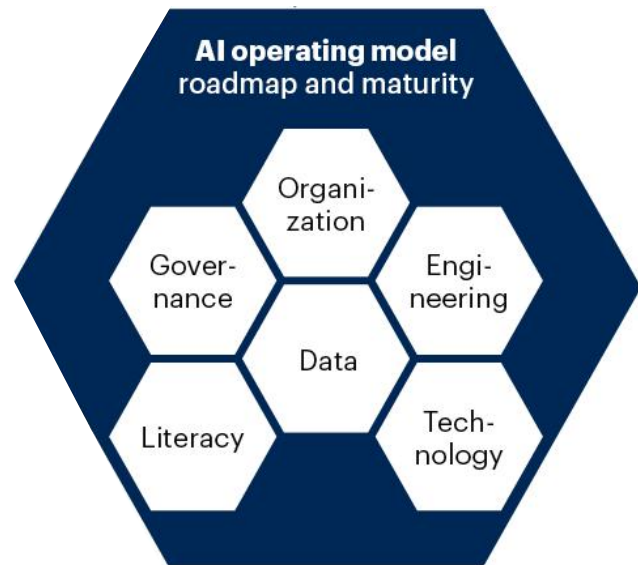
AI Operating Model (Part 2)



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The AI Operating Model in Context

- *Organisation*
- *Data*
- *Technology*
- *Engineering*
- *Governance*
- *Literacy*



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Technology

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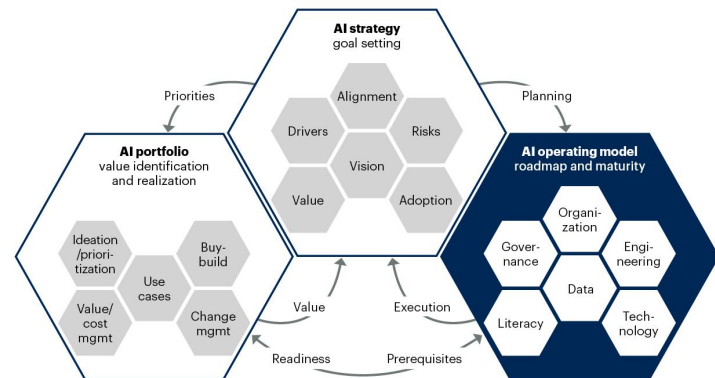
Technology (1)

What are the tools / platforms / infrastructure you will be working with?

Key considerations include:

- *Engineering: Integration with existing software and use cases. If customer-facing software runs on AWS, you should not deploy your customer-facing AI models on GCP.*
- *Literacy: how versed are teams in the use of a particular technology? Do they require up-skilling?*
- *Data: does your data work well with the tool? E.g. if you mainly use .csv files, then a columnar database like AWS Redshift may not be the best choice. Will you do streaming / near-realtime predictions, or just offline, batch predictions?*
- *Governance: some tools / platforms may not be compliant (yet).*

AI Operating Model as Part of the AI-Hive



Source: Gartner
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Bhati et al., 2024

Gartner

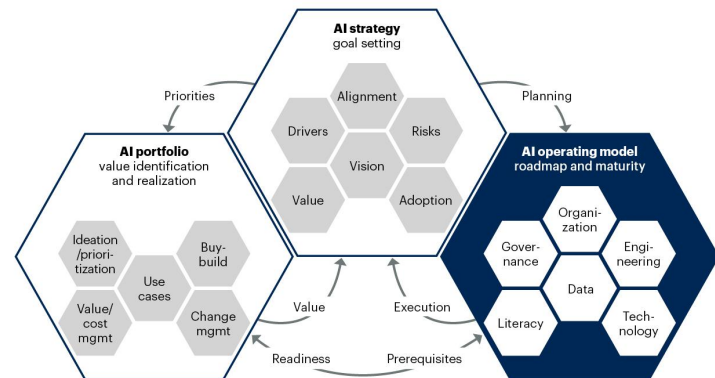
Technology (2)

What are the tools / platforms / infrastructure you will be working with?

Key considerations include:

- *Cost: Do you build it yourself or buy it off-the-shelf? How custom will your solutions be?*
- *Reliability: Does it matter if your AI application goes down for a few minutes / hours / days? What if data is lost?*
- *Scalability: Is it necessary to scale up to millions of users? In different regions? At varying workloads during the day? Can your AI API handle the traffic / tokens?*
- *Latency: How much response time do you have? Chatbots tend to have longer response times, whereas Image Generation is less forgiving (people will Google for pictures instead).*

AI Operating Model as Part of the AI-Hive



Source: Gartner
817549_C

Bhati et al., 2024

Gartner

Engineering

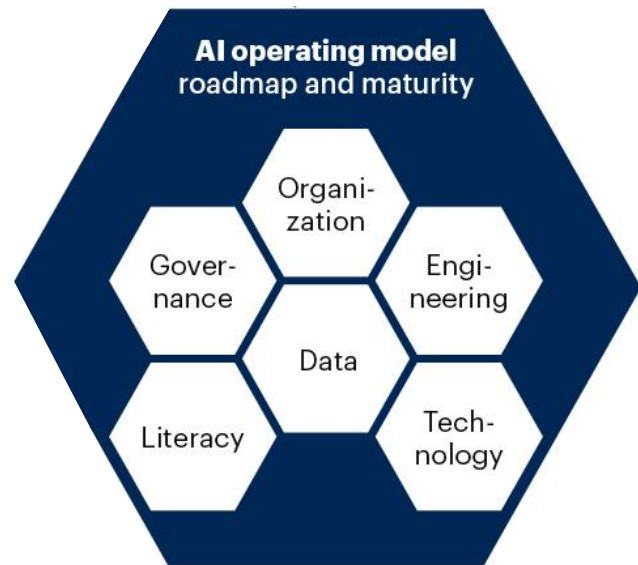


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Engineering

How you use the technology and data to achieve your goal

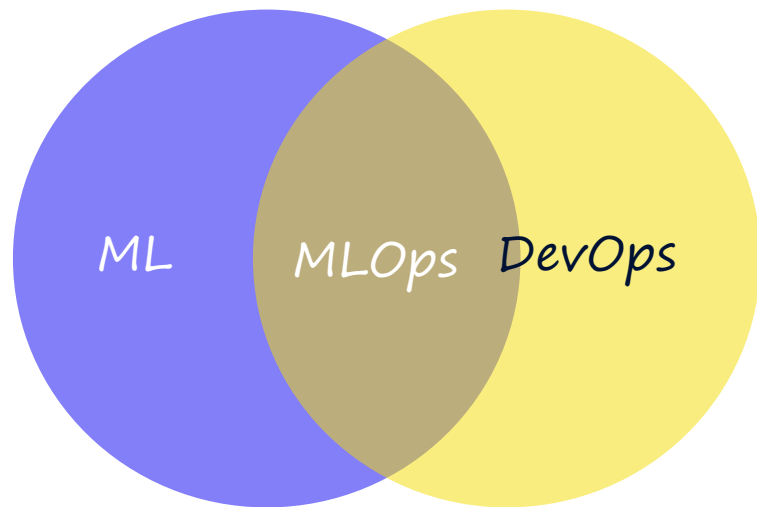
- *Data*
- *Technology*



ML(Ops) Engineering

Bridging the gap between AI and software development

- In theory: $MLOps = ML + DevOps$
- Role: In Team Topologies terms, this is an enabling role.
- Responsibilities: deploy models and pipelines in a reliable, secure, cost-efficient, and quick way.
- ML Engineering vs MLOps Engineering: Some companies may regard ML engineering as more ML heavy, and MLOps engineering as more Ops heavy, which makes it more of a platform role.



ML(Ops) Engineering skills

Bridging the gap between AI and software development

In practice:

- Data engineering: pre-process input and post-process output.
- ML: transform data scientist's code and deploy models.
- Software engineering: ML app development, software integration, APIs, scalability.
- DevOps engineering: manage infrastructure, reliability, cost-efficiency, monitoring.
- Security: secure your apps, data, models and pipelines.
- Governance: Ethics, Explainability

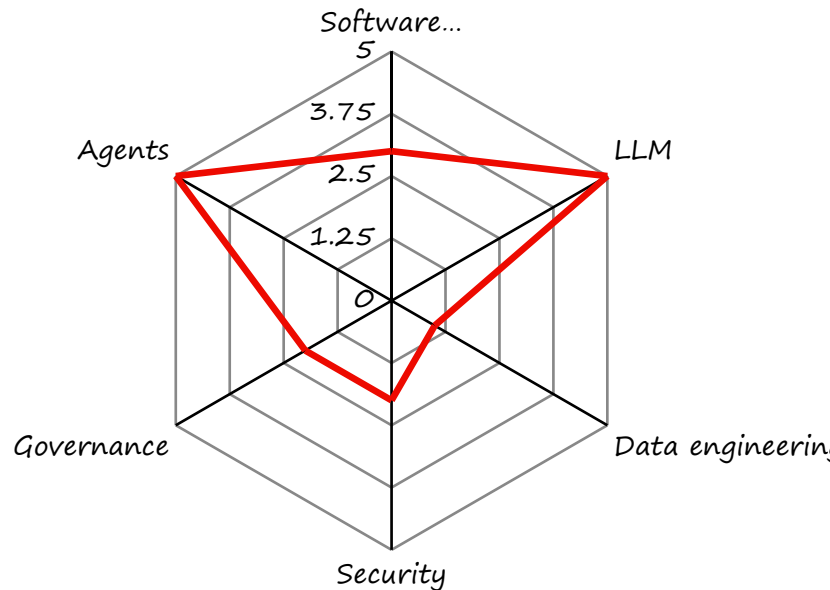


Image of a unicorn, MS Copilot

AI Engineering

More complex

- In theory: AI Eng = AI + Engineering
- Role: In Team Topologies terms, this is an enabling role.
- Responsibilities: pre-trains / fine-tunes models and deploys LLM and agents.
There is more focus on understanding of AI.



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Literacy

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Literacy

Multiple approaches to increase knowledge and skills

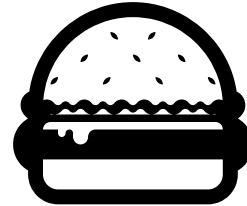
- Hire talent into teams.
 - Educate employees through training programs, meetups.
 - Provide project support through internal experts ('enabling team').
 - Organise Hackathons to share knowledge and increase enthusiasm.
 - Create guilds to share knowledge and skills (and give them mandate / projects).
-
- Mitigation option (will not increase literacy): Automate through your GenAI platform.

10-minute Break!

This will help you learn



10 minutes



"Teams need to be skilled enough to work with a tool" is an example of how technology interacts with:



A. Literacy

B. Data

C. Engineering

D. Organisation

How do latency requirements influence model selection? Select all that apply.



A. Higher latency means you can use a bigger model.

B. Lower latency means you can use a bigger model.

C. Higher latency means you are likely to use a smaller model.

D. Lower latency means you are likely to use a smaller model.

In terms of Team topologies, MLOps engineering is a/an _____ role



A. Stream-aligned

B. Enabling

C. Platform

D. Complicated sub-system



What is not a way to increase AI literacy?

A. Education

B. Hiring talent

C. Automate through your AI platform

D. Organise hackathons

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Data



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Data Requirements

To develop effective GenAI models, we may need:

- *Text Data: Large datasets of text samples from diverse sources (e.g., books, articles, conversations) to train language models.*
- *Image Data: High-resolution images with diverse labels and categories to train visual recognition models.*
- *Audio Data: Large collections of audio files with various formats (e.g., speech, music) to train speech-to-text and audio classification models.*
- *User Feedback Data: Real-time user feedback data to improve GenAI model performance and adaptability.*

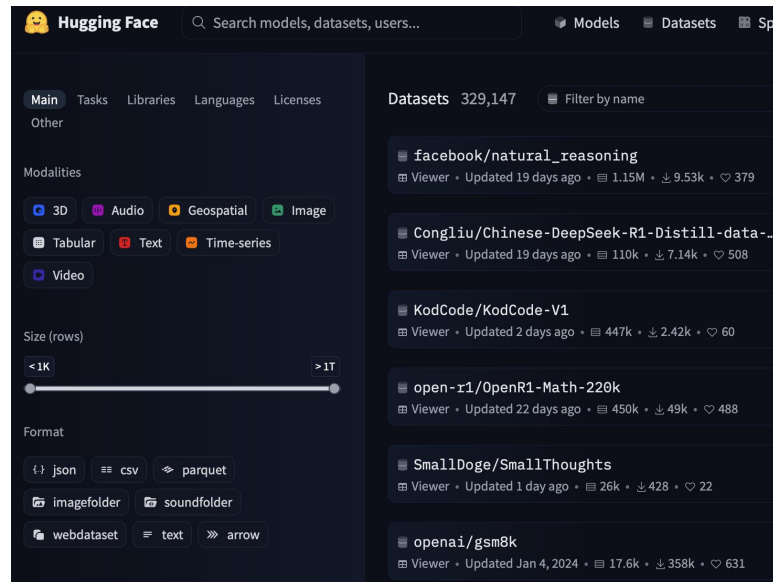


image about data, MS Copilot

Data Sources

Gather diverse and high-quality data

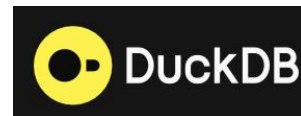
- *Public Datasets: Leverage publicly available datasets from reputable sources (e.g., Kaggle, Huggingface) for text, image, and audio data.*
- *Web Scraping: Utilise web scraping techniques to collect relevant data from websites, forums, and social media platforms.*
- *Crowdsourcing: Engage with platforms such as Amazon Mechanical Turk to collect labeled data.*
- *In-House Data Collection: Collect internal data from various sources (e.g., customer feedback forms, chat logs, customer interaction recordings).*
- *Partnerships and Collaborations: Collaborate with other organisations or research institutions to (buy and) access their datasets.*



Data Storage

To store large datasets efficiently

- *Cloud-based Storage: Utilise cloud storage services (e.g., Amazon S3, Google Cloud Storage) for scalable storage and access.*
- *Distributed File Systems: Leverage distributed file systems (e.g., HDFS) to handle massive dataset sizes.*
- *Data Lakes: Create data lakes using tools like Apache Spark or AWS Glue (with S3) to store and manage large datasets.*
- *Vector databases: Use vector databases like Weaviate, pgVector (PostgreSQL), DuckDB and Milvus to store images and text data.*



Data Storage Location within the Organisation

Key factors to consider:

- *Organisation: How are teams organised? Where are use cases owned? For example: if a use case is owned by a business team, then it may be wise to have domain data owned by them as well to ease data access.*
- *Literacy: Are people skilled enough to handle the data? Domain knowledge is often required, which means you may be limited if data is stored and processed centrally. On the contrary, business teams may not always be skilled enough to do data engineering.*

Data Processing

To ensure efficient processing and storage of data

- *Data Cleaning: Remove duplicates, handle missing values, and perform data normalisation.*
- *Feature Engineering: Extract relevant features from text, image, and audio data using techniques like bag-of-words, TF-IDF, or convolutional neural networks (CNNs).*
- *Data Augmentation: Apply data augmentation techniques to artificially increase dataset size and diversity.*
- *Data synthesis: Generate synthetic data to reduce the risk of data privacy breaches.*



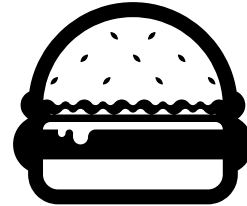
image of someone processing data, MS Copilot

7-minute Break!

This will help you learn



7 minutes



"Engage with platforms such as Amazon Mechanical Turk to collect labeled data." is an example of how data is collected through:



A. In-house data collection

B. Crowdsourcing

C. Web scraping

D. Partnerships and collaboration

Governance



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6 Risks of Language Models (Weidinger et al., 2021)

- *Discrimination, Exclusion and Toxicity: Social harms that arise from the language model producing discriminatory or exclusionary speech*
- *Information Hazards: Harms that arise from the language model leaking or inferring true sensitive information*
- *Misinformation Harms: Harms that arise from the language model providing false or misleading information*
- *Malicious Uses: Harms that arise from actors using the language model to intentionally cause harm*
- *Human-Computer Interaction Harms: Harms that arise from users overly trusting the language model, or treating it as human-like*
- *Automation, access, and environmental harms: Harms that arise from environmental or downstream economic impacts of the language model*

Risk Mitigation (Weidinger et al., 2021)

1. *Understanding the point of origin of a risk and its connections and similarities to other risks,*
2. *Identifying and implementing mitigation approaches,*
3. *The clear allocation of responsibility and implementation of corrective measures.*

Regulatory Frameworks for AI

A growing body of laws and guidelines

- *GDPR: The General Data Protection Regulation sets out strict rules for the processing of personal data in the EU. Note: European citizens are protected by GDPR.*
- *HIPAA: The Health Insurance Portability and Accountability Act regulates sensitive health information in the United States.*
- *Australian Privacy Act: This act protects individual privacy rights in Australia, including with regards to AI-powered systems.*
- *EU AI Act: Proposed by the European Commission, this act establishes a new regulatory framework for AI development and deployment*

Make sure to abide by these laws and guidelines when relevant

Australian Privacy Act (1988)

Who has rights under the Privacy Act?

The *Privacy Act 1988* regulates the way individuals' personal information is handled.

As an individual, the Privacy Act gives you greater control over the way that your personal information is handled. The Privacy Act allows you to:

- know why your personal information is being collected, how it will be used and who it will be disclosed to
- have the option of not identifying yourself, or of using a pseudonym in certain circumstances
- ask for access to your personal information (including your health information)
- stop receiving unwanted direct marketing
- ask for your personal information that is incorrect to be corrected
- make a complaint about an organisation or agency the Privacy Act covers, if you think they've mishandled your personal information.

<https://www.oaic.gov.au/privacy/privacy-legislation/the-privacy-act/rights-and-responsibilities>



image of data privacy, MS Copilo

Australian Privacy Act (1988)

Who has responsibilities under the Privacy Act?

Australian Government agencies (and the Norfolk Island administration) and organisations with an annual turnover more than \$3 million have responsibilities under the Privacy Act, subject to some exceptions.

What is an organisation?

The Privacy Act defines an 'organisation' as:

- an individual, including a sole trader (though generally, the Privacy Act doesn't apply to an individual acting in a personal capacity)
- a body corporate
- a partnership
- any other unincorporated association, or
- a trust.

unless they're a small business operator, registered political party, state or territory authority or a prescribed instrumentality of a state.

Some exceptions for business with turnover <\$3M

BUT

This is likely to change in 2025 due to the Privacy and Other Legislation Amendment Bill 2024

<https://www.oaic.gov.au/privacy/privacy-legislation/the-privacy-act/rights-and-responsibilities>

Australian Privacy Principles 2014

A summary for APP entities.

APP 01 Open and transparent management of personal information

Ensures that APP entities manage [personal information](#) in an open and transparent way. This includes having a clearly expressed and up to date APP [privacy policy](#).

APP 02 Anonymity and pseudonymity

Requires APP entities to give individuals the option of not identifying themselves, or of using a pseudonym. Limited exceptions apply.

APP 03 Collection of solicited personal information

Outlines when an APP entity can [collect](#) personal information that is solicited. It applies higher standards to the collection of [‘sensitive’ information](#).

APP 04 Dealing with unsolicited personal information

Outlines how APP entities must deal with unsolicited personal information.

APP 05 Notification of the collection of personal information

Outlines when and in what circumstances an APP entity that collects personal information must notify an individual of certain matters.

APP 06 Use or disclosure of personal information

Outlines the circumstances in which an APP entity may [use or disclose](#) personal information that it holds.

APP 07 Direct marketing

An organisation may only use or disclose personal information for [direct marketing](#) purposes if certain conditions are met.

APP 08 Cross-border disclosure of personal information

Outlines the steps an APP entity must take to protect personal information before it is [disclosed overseas](#).

APP 09 Adoption, use or disclosure of government related identifiers

Outlines the limited circumstances when an organisation may adopt a government related identifier of an individual as its own identifier, or use or disclose a government related identifier of an individual.

APP 10 Quality of personal information

An APP entity must take reasonable steps to ensure the personal information it collects is accurate, up to date and complete. An entity must also take reasonable steps to ensure the personal information it uses or discloses is accurate, up to date, complete and relevant, having regard to the purpose of the use or disclosure.

APP 11 Security of personal information

An APP entity must take reasonable steps to [protect](#) personal information it holds from misuse, interference and loss, and from unauthorised access, modification or disclosure. An entity has obligations to destroy or de-identify personal information in certain circumstances.

APP 12 Access to personal information

Outlines an APP entity's obligations when an individual requests to be given [access to personal information](#) held about them by the entity. This includes a requirement to provide access unless a specific exception applies.

APP 13 Correction of personal information

Outlines an APP entity's obligations in relation to [correcting the personal information](#) it holds about individuals.

EU Artificial Intelligence Act: Risk levels



"Harms that arise from the language model providing false or misleading information" is an example of the following Risk of Language Models:



A. Malicious Uses

B. Misinformation harms

C. Information hazards

D. Discrimination, Exclusion and Toxicity

According to the EU AI Act, social scoring is:



A. An unacceptable risk

B. A high risk

C. A medium risk

D. A low risk

Implementing Governance



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Data Governance

To ensure data quality and security

- *Data Quality: Regularly perform data quality checks to identify inconsistencies and errors.*
- *Data Encryption: Encrypt sensitive data at-rest, in-transit, and in-use. Encoding $\neq$ encryption.*
- *Identity and Access Management (IAM):
Implement access controls (e.g., role/attribute-based access control) to restrict data access to authorised personnel.*
- *Data Auditing: Conduct regular audits to ensure compliance with regulatory requirements.*

Model Governance

To ensure model quality, transparency and security

- *Transparency: Clearly define model development, deployment, and maintenance processes.*
 - *Document model architecture, algorithms, and data sources*
 - *Provide access to training data and testing protocols*
- *Accountability: Establish clear lines of responsibility for model development and deployment.*
 - *Identify key stakeholders and their roles*
 - *Define incident response procedures for errors or biases*
- *Fairness: Implement measures to detect and mitigate bias in AI models.*
 - *Use fairness metrics (e.g., bias detection)*
 - *Regularly audit and test for bias in data and model outputs*
- *Data Quality: Ensure high-quality data is used for training and testing AI models.*
 - *Establish data governance policies and procedures*
 - *Implement data validation and cleansing processes*
- *Security: Protect AI models from unauthorised access, tampering, or exploitation.*
 - *Use secure data storage and transmission practices*
 - *Implement access controls and authentication mechanisms*

Challenges for Model Governance

To ensure model quality, transparency and security

- *Data Quality Issues ('garbage in, garbage out'): Poor data quality can lead to biased or inaccurate AI models.*
- *Regulatory Complexity: AI model governance must comply with a wide range of regulations, including GDPR, CCPA, HIPAA, AI Act, etc.*
- *Technical Limitations: Limited technical capabilities can hinder the development and implementation of effective AI model governance frameworks.*
- *Scalability: As AI models become more complex and widespread, governance frameworks must be able to scale to meet growing demands.*
- *Stakeholder Management: Effective AI model governance requires managing diverse stakeholders with competing interests. For example, strict Model Governance (i.e. 'low risk appetite') may hinder innovation.*

Wrap-up



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Summary

- *The choice of Technology is very much dependent on Data and Literacy.*
- *Increase a company's Literacy by hiring talent, educating them and motivating them to join guilds, hackathons, etc.*
- *Ensure you have quality Data in place to train models, and store your data wisely.*
- *Governance ensures AI will be used responsibly and ethically.*
 - *The Australian Government is in the process of creating legislation around AI.*

For next lecture

*Advanced Topics and Emerging Trends in Business
Applications and Strategy*

References and Resources

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Questions



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